

## **Web Based Salon Management and Booking System**

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## Approval Sheet

*This design project entitled “**Project Title**” prepared by **Member A, Member B, Member C, and Member D** of the Computer Engineering Department, was examined and evaluated by the members of the Student Design Evaluation Panel and is hereby recommended for approval.*

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**SOFTWARE DESIGN PROJECT INFORMATION**

**1st Semester, SY 2025 - 2026**

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Student/Team</b><br><br><b>Team Mitochondria</b> | <b>Caasi, Karl Benedict</b><br><b>Mamanao, Kurt Marwin M.</b><br><b>Monoy, Justin Rhey</b><br><b>Poliño, Justine</b><br><b>Ramos, Jan Lawrence M.</b><br><b>Vasig, Yuan Hessed</b><br><b>Uy, Junichiro H.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Project Title</b>                                | <i>TITLE</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Project Concentration Area</b>                   | <i>Embedded Systems, Internet of Things, Machine Learning</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Design Objectives</b>                            | <p><i>The general objective is a general statement of the objectives of the project. This may contain the features that may be included in the design.</i></p> <p><i>Specifically, it aims to:</i></p> <ol style="list-style-type: none"> <li><i>1. Implementing an online booking platform to eliminate overlapping bookings, long phone queues, and missed opportunities caused by manual scheduling.</i></li> <li><i>2. Establishing a structured method for feedback collection via post-visit surveys to continuously improve service.</i></li> <li><i>3. Addressing marketing gaps and weak retention by consistent promotional campaigns.</i></li> <li><i>4. <b>Test and evaluate the system's accuracy.</b> This must remain for all projects; you can change "system" to the more appropriate word for the output (i.e. device).</i></li> </ol> |
| <b>Constraints</b>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Development Cost</b>                             | <i>Total build cost must not exceed ₱3,000</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Operational Cost</b>                             | <i>Yearly hosting and domain fees must stay within ₱2,200 – ₱4,200.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Page Load Speed</b>                              | <i>The website must load in under 3 seconds</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Mobile Responsiveness</b>                        | <i>Must achieve a 90% pass rate on Google Mobile-Friendly tests (essential for spa clients on phones)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                                                                                                         |                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scalability (Staff)</b>                                                                                                              | <i>The system is limited to a maximum of 3 staff accounts and 1 branch.</i>                                                                                                                                      |
| <b>Other constraints: These constraints do not affect each design; therefore, these were not included in selecting the best design.</b> |                                                                                                                                                                                                                  |
| <b>Manual Transaction Verification</b>                                                                                                  | <i>The system does not have an automated payment API (like PayPal). It is constrained to a manual "Upload Receipt" workflow where the staff must verify GCash screenshots before confirming the appointment.</i> |
| <b>Standards</b>                                                                                                                        |                                                                                                                                                                                                                  |
| <b>Sample Standard A</b>                                                                                                                | <i>Per constraint: define it and briefly explain how it was used in the project (See Appendix A, p. 60).</i>                                                                                                     |
| <b>Sample Standard B</b>                                                                                                                | <i>Per constraint: define it and briefly explain how it was used in the project (See Appendix B, pp. 61 - 62).</i>                                                                                               |

**For single pages, use "p." For multiple pages, use "pp."  
Appendices should be italicized and referred to every time it is mentioned.**

## **Abstract**

*This project addresses the need for enhanced comfort, security, and energy efficiency in residential environments by designing and implementing an intelligent home automation system leveraging Internet of Things (IoT) and machine learning techniques. The system integrates various smart sensors, actuators, and devices interconnected through a robust IoT framework. Machine learning algorithms enable the system to learn and adapt to residents' behaviors and preferences, providing personalized automation and control. Key features include automated lighting, climate control, security monitoring, and energy management. Performance evaluation through real-world scenarios demonstrates significant improvements in user convenience and energy savings. The project highlights the potential of combining IoT with machine learning to create an adaptive and efficient smart home environment, significantly enhancing the quality of life for residents by offering customized and efficient home automation solutions.*

*Keywords: Home Automation, Internet of Things (IoT), Machine Learning, Smart Sensors, Energy Efficiency, Security Monitoring, Adaptive Control, Smart Home*

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## List of abbreviation



## Definition of terms

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## CHAPTER 1: THE PROJECT AND ITS BACKGROUND

### 1.1 The Problem

Many small-to medium-scale salons continue to rely on manual and fragmented processes for managing daily operations such as appointment scheduling, customer records, billing, inventory monitoring, and marketing. These manual practices often result in overlapping bookings, missed or forgotten appointments, inaccurate transaction records, poor inventory tracking, and limited customer engagement. As customer demand increases and expectations shift toward digital convenience, these inefficiencies directly affect service quality, operational efficiency, and business profitability.

From an engineering design perspective, the core problem is the absence of an integrated, reliable, and scalable digital system that can automate salon operations while remaining simple enough for daily use by staff and clients. Without such a system, salon owners struggle to make data-driven decisions, customers experience inconvenience, and staff productivity is reduced. Addressing this problem requires the design and development of a web-based solution that transforms manual workflows into a centralized, automated, and user-friendly system.

### 1.2 The Client

The client for this project is **Claire Beauty Lounge**, a local salon business that provides beauty and personal care services to its customers. The salon currently operates using mostly manual procedures for booking appointments, managing customer information, tracking services rendered, and handling payments. The client seeks a digital solution that will improve operational efficiency, enhance customer experience, and strengthen the salon's online presence through a modern and accessible web-based system.

Table 1-1. Client and Engineering Requirements / Considerations

| Client Requirements / Considerations                    | Engineering Requirements / Considerations                                                     |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| The system can manage appointments efficiently          | The system must implement a reliable scheduling algorithm with real-time availability updates |
| The system can store and retrieve customer records      | The system must use a structured database with secure data storage and retrieval              |
| The system can generate billing and transaction records | The system must support accurate computation, record logging, and data consistency            |

### 1.3 The Project

*This project involves the design and development of a **Web Based Salon Management and Booking System** intended to automate and integrate the core operations of Claire Beauty Lounge. The system will provide features such as online appointment booking, customer relationship management (CRM), service and transaction tracking, inventory monitoring, and basic marketing support. By consolidating these functionalities into a single platform, the project aims to reduce manual workload, minimize operational errors, and improve overall service delivery.*

### 1.4 Project Objectives

*The general objective of this project is to design, develop, and evaluate a web-based salon management system that improves operational efficiency, customer satisfaction, and data organization for the client salon.*

*Specifically, the project aims to:*

1. **Design** the overall system architecture, including database structure and system workflows required for salon operations.
  - Design the data model for appointments, customers, services, and transactions.
2. **Develop** a web-based application that supports online booking, customer management, billing, and inventory tracking.
3. **Test and evaluate** the system's accuracy, reliability, and functional performance based on predefined requirements and real-world usage scenarios.

### 1.5 Scope and Delimitations

*The scope of this project includes the design and implementation of a web-based salon management and booking system tailored for Claire Beauty Lounge. The system will cover essential salon operations such as appointment scheduling, customer record management, service tracking, billing, and basic inventory monitoring. The target users of the system include salon owners, staff, and customers accessing the booking features. The project will be implemented and tested in a real salon setting to evaluate its effectiveness in improving workflow efficiency and customer experience.*

*The project is limited to the features and services identified during the requirements analysis phase. It does not account for variations in salon size, business models, or operational policies beyond the client's current setup. The system will only support the selected technology stack and will not integrate third-party platforms beyond those required for basic operation. Advanced analytics, full-scale marketing automation, and complex cybersecurity mechanisms are beyond the scope of this project. The system will provide a functional and basic user interface intended for demonstration and operational use rather than full commercial deployment.*

## 1.6 Design Constraints

### Usability

*Customers and salon employees should be able to easily operate the system without the need for technical expertise. To prevent errors when making reservations and making payments, clear menus and simple navigation are essential.*

### Accuracy

*Payments, services, and appointments must all be accurately documented. Accuracy is a crucial restriction since mistakes can result in multiple bookings, unhappy customers, and financial loss.*

### Dependability

*During business hours, there should be little downtime, and the system should function flawlessly. Reliable performance guarantees uninterrupted daily salon operations.*

### Cost

### Limitations

*For a small salon, the design must be reasonably priced. While still offering necessary features, development and maintenance expenses should be kept minimal.*

### Data

### Security

*Customer data needs to be kept confidential and secure. Sensitive data must be protected by basic security methods like login authentication.*

### Scalability

*Future expansion, such as the addition of more employees or services, should be supported by the system. Rebuilding the system later on is prevented by a flexible design.*

## 1.7 Engineering Standards

*The engineering standards provide the basis for the design, development, and evaluation of the proposed web-based salon management and booking system. Adherence to recognized software and information system standards ensures system reliability, data integrity, security, and maintainability. The project follows the applicable standards listed below.*

**ISO/IEC 25010:2011 – Systems and Software Quality Requirements and Evaluation (SQuaRE)**

*This standard defines quality characteristics for software systems, including functional suitability, reliability, usability, performance efficiency, security, maintainability, and scalability. The proposed system aligns with ISO/IEC 25010 to ensure that functional requirements such as appointment scheduling, data accuracy, and system reliability are met.*

### **ISO/IEC 27001 – Information Security Management**

*ISO/IEC 27001 provides guidelines for managing and protecting sensitive information. The system applies basic information security principles such as user authentication and controlled access to protect customer and transaction data.*

### **ISO/IEC 12207 – Software Life Cycle Processes**

*This standard defines processes for software development, testing, deployment, and maintenance. The project follows ISO/IEC 12207 to ensure that the system development lifecycle is systematic and properly documented.*

## **1.8 Engineering Design Process**

*The development of a web-based salon management and booking system requires a methodical approach to deal with operational problems like inefficient record management and manual appointment handling. A formal framework for determining system requirements, analyzing constraints, and creating practical alternatives is provided by an engineering-based methodology, called the Engineering Design Process. This ensures that the design constraints—such as usability, accuracy, reliability, cost, data security, and scalability—are in line with the objectives of the client and engineering requirements. The procedure makes it easier to make well-informed decisions regarding everything from problem identification to system improvement, while ensuring a reliable and functional solution.*



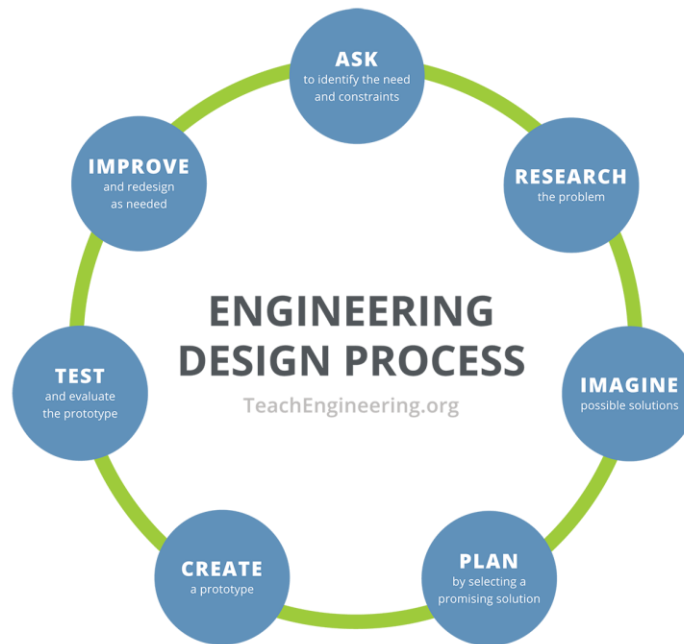


Figure 1.2 The Engineering Design Process (TeachEngineering, 2023)

Figure 1.2 illustrates the Engineering Design Process applied to the development of web-based salon management and booking systems. The process consists of a sequence of iterative stages that guide the identification of problems, development of solutions, and refinement of the final system.

### 1.8.1 Ask: Identifying the Need and Constraints

This stage focuses on identifying the problems and defining system requirements and limitations. For this project, the identified problems include manual appointment scheduling, inefficient customer record management, long waiting times, and overcrowding. Design constraints such as usability, accuracy, reliability, cost limitations, data security, and scalability are also established during this stage.

### 1.8.2 Research the Problem

The research stage involves gathering relevant information related to salon operations, booking workflows, and existing web-based management systems. This stage supports the identification of functional and technical requirements and provides a basis for informed design decisions.

### 1.8.3 Imagine: Develop Possible Solution

*In this stage, possible solutions to the identified problems are generated. These include the development of a centralized web-based system capable of handling appointment scheduling, customer records, billing, and inventory management. Multiple system features and design approaches are considered.*

#### **1.8.4 Plan: Select a Promising Solution**

*The plan stage involves selecting the most feasible solution based on the defined requirements and constraints. For this project, a web-based salon management and booking system is selected as the proposed solution. System architecture, database structure, and process workflows are defined during this stage to ensure consistency, reliability, and scalability.*

#### **1.8.5 Create: Build a Prototype**

*This stage involves the development of a functional prototype of the selected system. Core features such as online appointment booking, customer information management, transaction recording, and inventory tracking are implemented according to the system plan.*

#### **1.8.6 Test and Evaluate the Prototype**

*The testing stage evaluates the system's functionality, accuracy, and reliability. The system is tested against predefined requirements to verify correct scheduling behavior, data consistency, and stable performance during normal operation.*

#### **1.8.7 Improve: Redesign as Needed**

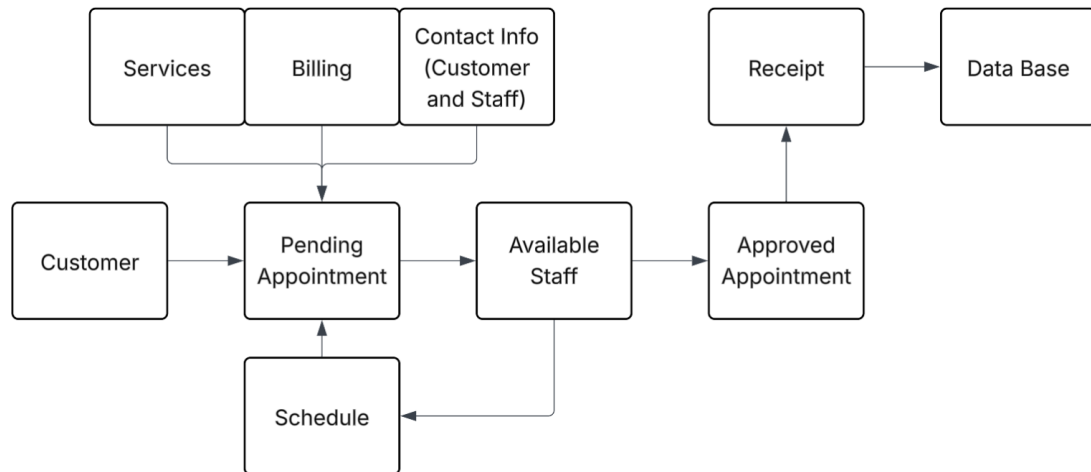
*The improvement stage focuses on refining the system based on testing results and evaluation outcomes. Necessary modifications are applied to address identified limitations and enhance system performance. This iterative process continues until the system meets the defined requirements.*

## CHAPTER 2: SOFTWARE DESIGN

<Brief description of chapter 2>

### 2.1 Description of the Design Solution

#### 2.1.1 General Description



*Figure 1 Illustrative Diagram of the System*

*The basic function of our web-based salon appointment system starts with the customer. The customer booked an appointment; this pending appointment contains the services along with its prices; the customer can also choose the preferred time. Once this pending appointment reaches the available staff, they can both communicate and agree on a fixed time through the email. Once the appointment is done, the staff is the only one who can approve and mark it as finished, and this will generate the receipt and will put this transaction record in the database for history viewing.*

#### 2.1.2 Engineering Principles Involved

*Short introduction of the section.*

##### **Engineering Principle**

*Explanation of relevant studies to the principles used.*

#### 2.1.3 Prior Art Analysis

*Discusses the existing solutions with existing patents (and other similar registrations) and compare them with the design project.*

*Features of existing solutions that are particularly of interest should be discussed. Not necessarily aligned completely with the title; what sets your project apart?*

*Matrix Format is preferred, in addition to narratives. Explain the table.*

*Table xx Prior Art Analysis Matrix*

| Design      | Features  |           |           |           |           |
|-------------|-----------|-----------|-----------|-----------|-----------|
|             | Feature A | Feature B | Feature C | Feature D | Feature E |
| Prior Art A | X         |           | X         |           |           |
| ...         |           |           |           |           |           |
| Prior Art n |           | X         |           |           |           |
| PROJECT     | X         | X         | X         | X         | X         |

## 2.2 General System Architecture

*This section outlines the implementation of the engineering concepts and elements described earlier, focusing on the overall architecture, hardware, and software components of the salon automation system. The system is primarily web-based, with minimal hardware components required.*

### 2.2.1 Hardware Elements

*Given that the system is a web-based solution, the hardware requirements are minimal and mainly support the user interfaces and backend services:*

- **Servers:** *The backend of the salon system is hosted on cloud-based servers, such as those provided by AWS, Google Cloud, or Azure. These servers handle the web application, database storage, and API services.*
- **User Devices:** *Customers and salon staff interact with the system through various devices, including smartphones, tablets, and desktop computers. These devices provide access to the web-based platform, allowing for appointment booking, reminders, and payment processing.*

### 2.2.2 Software Elements

*The software architecture consists of multiple layers, with various programming languages, frameworks, and tools used to develop the system's frontend, backend, and third-party integrations.*

**A. Embedded Software**  
*There is no embedded software required for this system, as the system operates entirely through web interfaces on user devices and cloud services.*

#### B. Application Software

- **Frontend:** The user interface is developed using HTML, CSS, and JavaScript, with frameworks such as React for a dynamic, responsive design. The platform is optimized for desktop and mobile devices, with potential use of frameworks like React Native or Flutter for mobile app development.
- **Backend:** The backend is powered by technologies such as Node.js (JavaScript), Django (Python), with a database system such as PostgreSQL or MySQL to store client records, appointment information, inventory, and billing data.
- **CRM System:** The platform integrates with a Customer Relationship Management (CRM) system to store customer data, including visit history, preferences, allergies, and stylist preferences.
- **API Integrations:** The system uses third-party APIs for sending automated notifications (via Twilio for SMS, Mailchimp for email campaigns) and managing social media interactions (Instagram/Facebook integrations).

### C. Key Algorithms Used

- **Appointment Scheduling Algorithm:** An algorithm that checks stylist availability and prevents double-booking. It also optimizes scheduling to maximize stylist utilization.
- **Inventory Management Algorithm:** This algorithm tracks product usage and triggers reorder alerts when inventory levels fall below predefined thresholds.
- **Customer Retention Algorithm:** The system uses predictive analytics to recommend loyalty offers and rebooking reminders based on customer behavior and preferences.

### 2.2.3 System Algorithm

The salon automation system relies on several core algorithms to ensure smooth operations:

- **Appointment Scheduling:** When a customer books an appointment, the system verifies stylist availability, considers service duration, and schedules the appointment accordingly. It also sends automated reminders to the customer and stylist.
- **Notifications:** Automated reminders are sent via email and SMS to customers before appointments. In cases of cancellations, the system applies cancellation policies and updates availability in real-time.
- **Inventory Management:** The inventory management algorithm tracks product usage and sends alerts when stock levels are low, enabling the salon to reorder products before they run out.
- **Payment Processing:** Payments are securely processed through integrated payment gateways, and digital receipts are issued immediately. The system also manages loyalty points and promotional discounts during the checkout process.

### 2.2.4 Data, Datasets, and Processing

#### a. Datasets

The system utilizes several datasets, including:

- **Customer Data:** Information such as customer names, contact details, service preferences, and visit history.
- **Appointment Data:** Data related to booking details, such as service type, stylist assigned, and appointment times.
- **Inventory Data:** Information about products used in the salon, including quantities, product names, and reorder levels.
- **Staff Data:** Details about staff availability, roles, and shifts.

*Data acquisition methods include:*

- **Customer Input:** Customers provide personal details and preferences when they register or book appointments.
- **External APIs:** Data from external services (e.g., social media platforms for marketing, or SMS gateways for notifications) is collected through API integrations.
- **Internal Logs:** Data generated by the system, including appointment logs, customer interactions, and inventory updates, is stored in the system database.

#### **b. Data Processing Scheme and Algorithms**

- **Pre-processing:** Raw customer data is validated and formatted. Appointment data is checked for availability, and stylist schedules are updated in real-time. Inventory data is checked for accuracy before updates are made.
- **Post-processing:** After a customer book an appointment, the system sends automated confirmation emails and SMS reminders. Inventory is updated following each service to reflect the consumption of products.

*Example of dataset processing:*

- **Raw Dataset:** Customer name, preferred stylist, and appointment time.
- **Processed Dataset:** The system confirms stylist availability, updates the calendar, and sends confirmation to the customer.

#### **c. Other Data Utilized in the Design**

- **Mock Data:** During testing and development, mock data is used to simulate customer interactions, appointments, and payment processing to ensure the system functions correctly before going live.
- **Business Rules Data:** This includes rules for appointment cancellations, loyalty program points of accumulation, and promotional discounts. These rules help automate client retention and marketing efforts.

## **2.3 Design Alternatives**

### **2.3.1 Rationale for Design Alternatives**

*The design alternatives presented in this section were selected to address the core problem of automating salon operations while considering feasibility, scalability, cost, and technical complexity. These alternatives represent realistic and commonly adopted system architectures for small-to-scale business applications. Each design alternative explores a different level of system integration and technological sophistication, allowing comparison in terms of performance, maintainability, and alignment with client requirements.*

*These design alternatives are critical to the engineering design process because they provide structured options before committing to a final solution. By evaluating multiple approaches, the project minimizes design bias, avoids premature optimization, and ensures that the selected design offers the best balance between functionality and implementation constraints.*

#### **2.3.1 Design Alternative 1:**

##### **A. Engineering Principles of Alternative**

*Contains discussions (with references) of the technologies, principles, and concepts utilized (i.e. Machine Learning, Convolution, Kinematics, etc.).*

*Note: This repeats for all 3 designs. It must not mention the principles mentioned above but instead those specific to the design alternative.*

##### **B. Architecture of Design Alternative**

*Discuss how key components, sub-systems, algorithms, and alternatives of this design are implemented.*

##### **C. Constraints**

*Constraint A*

*Constraint B*

*Constraint C*

*Constraint D*

*Constraint E*

##### **iv. Evaluation Results (if model)**

### 2.3.2 Design Alternative B

*i. Engineering Principles of Alternative*

*ii. Architecture of Design Alternative*

*iii. Evaluation of Constraints*

**Constraint A**

**Constraint B**

**Constraint C**

### 2.3.3 Design Alternative C

*i. Engineering Principles of Alternative*

*ii. Architecture of Design Alternative*

*iii. Constraints*

**Constraint A**

**Constraint B**

**Constraint C**

## 2.4 Standards Involved in the Design

*This section presents the standards followed by the design, including their references. Matrix may be used to show how standards are used in each specific design.*

*Table xx Summary of Standards Involved in the Alternatives*

| Standard | Brief Description | DESIGNS  |          |          |
|----------|-------------------|----------|----------|----------|
|          |                   | DESIGN A | DESIGN B | DESIGN C |



|                                                                 |                                                                                             |                                                                              |           |                                                               |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------|---------------------------------------------------------------|
| <i>IEC 60950</i>                                                | <i>Product Safety Standard for electronic and computing products.</i>                       | <i>Used in enclosure, power supply leakage, ESD, wiring, and connectors.</i> |           |                                                               |
| <i>Philippine National Standards for Drinking Water (PNSDW)</i> | <i>Standards for drinking water quality, water sampling and examination and evaluation.</i> | <i>Used in conditional statements to determine if water is drinkable.</i>    |           |                                                               |
| <i>IEEE 1309-2013</i>                                           | <i>Standard for Calibration of Electromagnetic Field Sensors and Probes</i>                 | <i>NA</i>                                                                    | <i>NA</i> | <i>Calibration of sensors used in detecting heavy metals.</i> |
| <i>IEEE 1858-2016</i>                                           | <i>IEEE Standard for Camera Phone Image Quality</i>                                         | <i>Reference for image processing cameras.</i>                               | <i>NA</i> | <i>NA</i>                                                     |

*Explain this table and end with a summary.*

## CHAPTER 3: DESIGN TRADEOFFS

### 3.1 Summary of Constraints

*Explain table xx below in this paragraph.*

*Table xx Summary of Design Constraints*

| Designs         | Constraints              |                          |                          |                          |                          |
|-----------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
|                 | Constraint A<br>(Metric) | Constraint B<br>(Metric) | Constraint C<br>(Metric) | Constraint D<br>(Metric) | Constraint E<br>(Metric) |
| <i>Design A</i> |                          |                          |                          |                          |                          |
| <i>Design B</i> |                          |                          |                          |                          |                          |
| <i>Design C</i> |                          |                          |                          |                          |                          |

*Synthesize for the next section.*

### 3.2 Trade-offs

*Table xx Preference and Importance of Constraints*

| Constraints | Preference | Importance (raw) | % Importance |
|-------------|------------|------------------|--------------|
|             |            |                  |              |

*Explain the use of Pareto Multi-Criteria Decision Making (MCDM).*

$$\text{Minimization} = 9 \times \left( \frac{\text{Max Value} - \text{Raw Value}}{\text{Max Value} - \text{Min Value}} \right) + 1 \text{ Equation No. xx}$$

$$\text{Maximization} = 9 \times \left( \frac{\text{Raw Value} - \text{Min Value}}{\text{Max Value} - \text{Min Value}} \right) + 1 \text{ Equation No. xx}$$

### 3.2.1 Tradeoff 1: Constraint A (Metric)

#### 3.2.1.1 Design 1: Normalization of Constraint A (Metric)

<Introduce>

*Table xx Evaluation of Three Design Alternatives based on Constraint A*

| Design | Constraint (Metric) |
|--------|---------------------|
|        |                     |
|        |                     |
|        |                     |

<Analyze>

#### 3.2.1.2 Design 2: Normalization of Constraint A (Metric)

*Table xx Evaluation of Three Design Alternatives based on Constraint B*

| Design | Constraint (Metric) |
|--------|---------------------|
|        |                     |
|        |                     |
|        |                     |

#### 3.2.1.3 Design 3: Normalization of Constraint A (Metric)

*Table xx Evaluation of Three Design Alternatives based on Constraint C*

| Design | Constraint (Metric) |
|--------|---------------------|
|        |                     |
|        |                     |
|        |                     |

### 3.2.2 Tradeoff 2: Constraint B (Metric)

#### 3.2.2.1 Design 1: Normalization of Constraint B (Metric)

#### 3.2.2.2 Design 2: Normalization of Constraint B (Metric)

#### 3.2.2.3 Design 3: Normalization of Constraint B (Metric)

### 3.2.3 Tradeoff 3: Constraint C (Metric)

3.2.3.1 Design 1: Normalization of Constraint C (Metric)

3.2.3.2 Design 2: Normalization of Constraint C (Metric)

3.2.3.3 Design 3: Normalization of Constraint C (Metric)

### 3.2.4 Tradeoff 4: Constraint D (Metric)

3.2.4.1 Design 1: Normalization of Constraint D (Metric)

3.2.4.2 Design 2: Normalization of Constraint D (Metric)

3.2.4.3 Design 3: Normalization of Constraint D (Metric)

### 3.2.5 Tradeoff 5: Constraint E (Metric)

3.2.5.1 Design 1: Normalization of Constraint E (Metric)

3.2.5.2 Design 2: Normalization of Constraint E (Metric)

3.2.5.3 Design 3: Normalization of Constraint E (Metric)

## 3.3 Summary of the Normalized Values of the Three Designs

| Designs  | Constraints              |                          |                          |                          |                          |
|----------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
|          | Constraint A<br>(metric) | Constraint B<br>(metric) | Constraint C<br>(metric) | Constraint D<br>(metric) | Constraint E<br>(metric) |
| Design A |                          |                          |                          |                          |                          |
| Design B |                          |                          |                          |                          |                          |
| Design C |                          |                          |                          |                          |                          |

## 3.4 Designers Raw Ranking for the Three Designs

Table xx Designers Raw Ranking for the Three Designs

| Decision<br>Criteria | Criterion's Importance |                   | Ability to Satisfy Criterion |          |          |
|----------------------|------------------------|-------------------|------------------------------|----------|----------|
|                      | Scale (0-10)           | Percentage<br>(%) | Design A                     | Design B | Design C |
|                      |                        |                   |                              |          |          |
|                      |                        |                   |                              |          |          |
|                      |                        |                   |                              |          |          |
|                      |                        |                   |                              |          |          |
|                      |                        |                   |                              |          |          |

|  |  |  |  |  |  |
|--|--|--|--|--|--|
|  |  |  |  |  |  |
|--|--|--|--|--|--|

### **3.5 Sensitivity Analysis**

### **3.6 Influence of the Design Tradeoffs in the Final Design**

## CHAPTER 4: FINAL DESIGN

### 4.1 Final Design

#### 4.1.1 Software Application

#### 4.1.2 Hardware Design

### 4.2 Test Procedures and Evaluation

#### 4.2.1 Test Procedures

#### 4.2.2 Test Evaluation

### 4.3 Test and Evaluation Results

#### 4.3.1 Test Results

#### 4.3.2 Evaluation Results

### 4.4 Conclusion

### 4.5 Impact of the Design

#### 4.5.1 Societal

*Target UN SDG.*

#### 4.5.2 Ethical

*In compliance with known ethical codes/standards*

#### 4.5.3 Legal

*National / Intl Laws*

### 4.6 Sustainability Plan

## **CHAPTER 5: BUSINESS PLAN AND MODEL**

### **5.1 Business Plan**

#### **5.1.1 Executive Summary**

#### **5.1.2 General Company Description**

#### **5.1.3 Products and Services Offered**

#### **5.1.4 Marketing Plan**

#### **5.1.5 Marketing Strategy**

### **5.2 Business Model**

### **5.3 Intellectual Property (IP) Reports**

## REFERENCES

Note: This must be done using APA format. Check the guide for more details:  
<https://www.scribbr.com/apa-style/apa-seventh-edition-changes/>

Covey, S. R. (2013). *The 7 habits of highly effective people: Powerful lessons in personal change*. Simon & Schuster.



## APPENDICES

*Include standards preview, certification from experts/clients, code snippets, patent reports, and other long and detailed documents here. Format is as follows below:*

### **APPENDIX A: TITLE OF THE SECTION**

*<figure>*

*Note: No figure number. Standards must be followed with a paragraph explaining its contents and purpose.*